

Q1. Create a DataFrame containing student performance data with attributes such as Study Hours, Attendance, Assignment Score, and Exam Score. Calculate the correlation matrix for these attributes and generate a Heat Map to visualize the correlation between them. Display the correlation values, suitable labels, and an appropriate title.

[10 Marks]

Q2. Write a Python program to create a DataFrame with the attributes Temperature, Humidity, Wind-Speed, Pressure, and Rainfall. Apply the following data transformation techniques and display the transformed data:

- a)Min-MaxScaling
- b)Standardization
- c) Normalization

[20 Marks]

OR

Q2. Implement the Apriori algorithm on the given student activity dataset to identify frequent itemsets and generate association rules among the activities. Repeat the analysis using minimum support values 0.2 and 0.3.

[20 Marks]

TID	Items
1.	Library, Seminar
2.	Workshop, Project
3.	Library, Assignment
4.	Seminar, Workshop
5.	Assignment, Seminar
6.	Workshop, Assignment
7.	Project, Library
8.	Library, Seminar, Project
9.	Library,Seminar,Workshop

Q3. Viva

[5 Marks]